

Venkatesh Kumar Raju

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Open to Relocate | [LinkedIn](#) | [GitHub](#)

SUMMARY

I am a dedicated GIS Engineer, a DGCA Certified Remote Pilot, and an OpenCV Master with a strong background in deep learning using Pytorch, Keras, and TensorFlow. My experience in geospatial analysis, 3D mapping, drone operations, and computer vision. I have applied my skills to aerial imagery projects, digital elevation models, and 3D mapping. I am a team worker with strong leadership skills and the ability to adapt to new situations. I am always eager to learn and apply new technologies to solve complex problems.

WORK EXPERIENCE

SeaMount Geospatial LLC

Geographic Information System Developer and Machine Learning Engineer, Brooklyn, New York (Remote) Oct 2025 — Present

Junior GIS Specialist & Machine Learning Engineer, Brooklyn, New York (Remote) Oct 2024 — Oct 2025

Led the development of CORAL INTEL, a powerful deep learning platform designed to support marine scientists, reef managers, and citizen explorers in monitoring and analyzing coral reef ecosystems

- Key Contributions:

-  Data Acquisition & Preparation: Curated a high-quality dataset of 10,000+ marine images from public repositories, including GBIF (Global Biodiversity Information Facility). Ensured diverse and representative coverage across species and reef environments.
-  Annotation Pipeline Management: Managed the annotation workflow using Roboflow and labeling, creating detailed guidelines and quality control checks for both bounding boxes and segmentation masks, resulting in consistent and accurate labels.
-  YOLO Model Implementation (v8, v9, v11, v12, YOLOe): Developed and fine-tuned multiple YOLO-based models in PyTorch, applying transfer learning and data augmentation strategies. Achieved an mAP50-95 of 0.86% on the object detection task and an mAP50-95 of 0.807 % on the segmentation task.
-  Web Application Development (Flask): Designed and built a user-friendly Flask web interface to visualize real-time object detection and segmentation results. Implemented APIs to support:
 - Remote reef monitoring and image upload
 - Interactive visualization and annotation review
 - Seamless frontend-backend communication

MAPe IT SOLUTIONS

Internship, Chennai, India

Nov 2022

GIS digitization and mapping of solar panels, swimming pools, and building structures.

Anna University

Placement Representative, Chennai, India

Aug 2021 — Mar 2023

As a Placement Representative, I served as the vital link between the placement cell and our student community. My responsibilities included disseminating information about job opportunities, internship programs, and recruitment drives. I actively communicated with students, addressing their queries, clarifying procedures, and guiding them through the placement process.

Geo Adithya Technologies

Junior Photogrammetrist Engineer, Chennai, India

Mar 2019 — Dec 2019

Digitization Specialist: Responsible for creating planimetric mappings, digital surface models, and digital terrain models, ensuring quality control throughout the process.

SKILLS

Geospatial	: Photogrammetry, DTM, DSM, GCP, Contour, Topography, Ortho map, 3D Mapping, Coordinate Systems
Image Processing	: Image Processing, Computer Vision, OpenCV, Image Classification
Data Management	: Data Processing, Data Analysis, Interpreting Data, Asset Management, Spatial Data
Software Proficiency	: Drone2map, webODM, Dronelink, AutoCAD, ArcGIS Pro, ArcGIS, QGIS, AutoCAD Map 3D
Programming	: Python, MATLAB
Machine learning	: PyTorch, TensorFlow, Keras
App Development	: Flask

Interests	: Drone Piloting , Computer Vision, Machine Learning , Photogrammetry , GeoAI , Object Detections
Languages	: Tamil, English, French level-A2

EDUCATION

College of Engineering, Guindy, Chennai, India - Master, Remote Sensing and Geomatics - (GPA: GPA: 7.2)

During my master's program, I embarked on a fascinating project that combined the power of a hedonic model a method commonly used in real estate economics and artificial neural networks. Specifically, I explored how various features (such as location, amenities, and land characteristics) impact land prices in the Paruthipattu. By leveraging geospatial data and machine learning techniques, I aimed to create accurate predictions for land values. This interdisciplinary approach allowed me to bridge the gap between traditional statistical modeling and cutting-edge neural networks.

S.A Engineering college, Chennai, India - Bachelor, Civil Engineering - (GPA: GPA: 6.5)

During my undergraduate studies, I immersed myself in the fascinating field of civil engineering. I gained foundational knowledge in areas such as surveying and infrastructure development. The curriculum allowed me to explore the principles behind designing and constructing essential structures that shape our world. Whether it was analyzing structural stability or understanding materials science, my time at S.A Engineering College laid the groundwork for my career in civil engineering.

CERTIFICATIONS

Remote Pilot Certificate, DGCA

Jan 2024

CERTIFICATIONS LINK

PROJECTS

Image Stitching , [Link](#)

- Stitch drone images together to create a panoramic orthomosaic.

Converting JPG to GeoTIFF, [Link](#)

- Converting JPG images with embedded GPS information into georeferenced GeoTIFF files.

Virtual-Makeup, [Link](#)

- Project demonstrates facial landmark detection and the implementation of various features using OpenCV, Dlib, and Matplotlib.

Object Detection, [Link](#)

- In this project Object detection is done using Yolov5 . ANPR (Automatic Number Plate Recognition) Detecting the number plate is a prerequisite to solving the ANPR problem.

CORAL INTEL, SeaMount Geospatial Labs, [Link](#)

- CORAL INTEL, a powerful deep learning platform designed to support marine scientists, reef managers, and citizen explorers in monitoring and analyzing coral reef ecosystems